Using Mutil-Agentic AI (LLMs) to Generate Adversarial Attacks on Other ML Models

ECE 653- Scholarly Paper Presentation

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Problem Statement

Current Adversarial attacks:

- Static Adversarial attacks fail against adaptive defenses
- Manual red-teaming is time consuming and resource intensive.
- No unified framework exists for testing multiple model types.

Our Solution:

- LLM-powered agent that autonomously discovers vulnerabilities and executes an attack
- Adapts attack strategies based on previous failures
- A framework that can support vision, audio, tabular, and LLM targets.

Idea: from C & W paper about successfully performing attacks on all types of models at the time of their paper.

We use LLMs as a guide or “thinker” to make successful attacks on any model

Comparison with Prior Work

1] Chao et al. (2024) - Jailbreaking LLMs in 20 Queries

- Our work EXTENDS this: same iterative approach
- But applies to vision/audio/tabular, not just LLMs

[2] Zhou et al. (2025) - AutoRedTeam

- Our work BUILDS ON THIS: memory-based attack integration
- But adds multi-modal support + ART toolkit integration

[3] Carlini & Wagner (2017) - CW Attack

- Our work USES THIS: as an execution tool
- But automates parameter selection instead of manual

Key Differentiator:

No prior work combines ALL of:

- LLM-driven planning
- Multi-agent attack execution
- Transparent reasoning logs

Agentic Framework: Approach

Initially:

Started with LangChain, but due to cyclic needs for the execution workflow.

CURRENT:

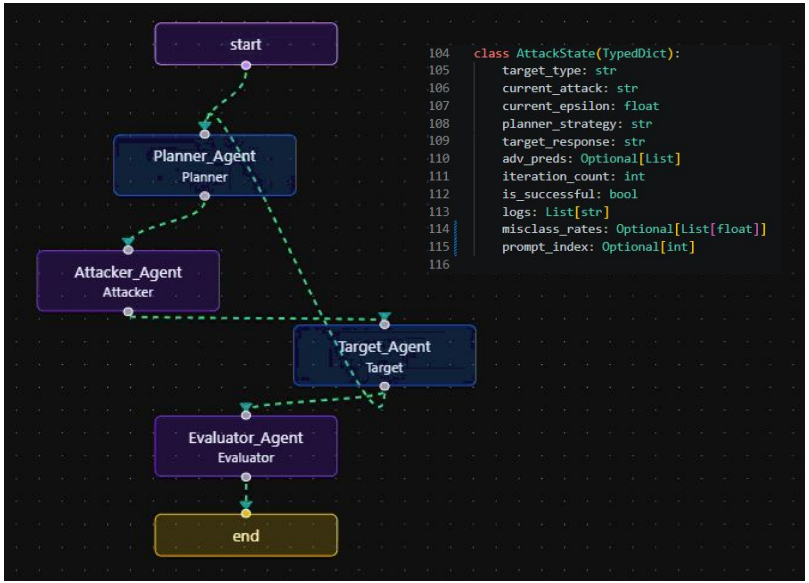
Used LangGraph to generate the framework.

Planner Chooses attack type & tunes the hyper parameters based on history

Attacker Executes the Attacks with the ART toolkit provided

Target Vision/LLM model being attacked

Evaluator Check for success rates and metrics



Prior to an any Attack: Model and Data Preprocessing

Prior to an attack we developed dedicated scripts for each use case:

- vision_target.py
- audio_target.py
- tabular_target.py
- llm_target.py

Followed the same workflow:

- Standardize a Model Load, and Model Train, which provided a plug-and-play, use case of the Agentic Workflow.
- Also made sure the Target Model was compatible with the ART toolkit.
- All these files followed one specific structure, where we loaded the model, and the dataset we want to test the models against, and after loading it, try to train the models and then return the trained model so that the target node in the Agentic Framework can perform a successful attack.

Experimental Setup

- **Vision Whitebox Attacks:** FGSM, PGD, DeepFool, Carlini L2, Adversarial Patch
- **Vision Blackbox Attacks:** hopskipjump, ZOO, square, simba, boundary
- **Audio Attacks:** CW-ASR, Imperceptible-ASR, Boundary(Stubbed)
- **Tabular Attacks:** ZOO, boundary, DecisionTree
- **Supported Architectures:** ResNet50, ResNet50_AT, ViT, ViT_AT, Wide_ResNet, DenseNet, EfficientNet, Convnext

- **Datasets Used:** ImageNet, Speech Command, Iris

All attacks are iterative and will be logged until a threshold T for misclassification rate M is met (fixed to M > 98%).

Logs: Example

List of Agents

Chain of thought reasoning for each agent

```
[Setup] Clean accuracy baseline: 80.0%
[PLANNER | Iteration 0]:
Since this is the first iteration, I'll start with a simple and fast attack to get an initial understanding of the model's vulnerabilities.

The 'fgsm' (Fast Gradient Sign Method) attack is a good starting point as it's relatively quick to compute and requires minimal hyperparameter tur

I'll set epsilon to 0.1, which is a common value for initial experiments. This will introduce a noticeable perturbation while still allowing the m

ATTACK: fgsm
EPSILON: 0.1
→ Chose: fgsm, epsilon=0.1
[ATTACKER | Iteration 0]:
Attack: fgsm | Epsilon: 0.1
Result summary: Attack 'fgsm' (eps=0.1) generated 100 adversarial examples. Sample adv predictions (first 5): [955, 489, 911, 851, 611] | Clean pr
[TARGET | Iteration 0]: Vision attack already executed in attacker_node.
[EVALUATOR | Iteration 0]:
Misclassification rate: 93.0% (93/100)
Verdict: FAILURE
Reasoning: Attack 'fgsm' (eps=0.1) misclassified 93/100 images (93.0%). Below 0.98% threshold — will escalate attack next iteration.
[PLANNER | Iteration 1]:
```

```
ATTACK: fgsm
EPSILON: 0.1
→ Chose: fgsm, epsilon=0.1
[ATTACKER | Iteration 0]:
Attack: fgsm | Epsilon: 0.1
Result summary: Attack 'fgsm' (eps=0.1) generated 100 adversarial examples. Sample adv predictions (first 5): [955, 489, 911, 851, 611] | Clean predi
[TARGET | Iteration 0]: Vision attack already executed in attacker_node.
[EVALUATOR | Iteration 0]:
Misclassification rate: 93.0% (93/100)
Verdict: FAILURE
Reasoning: Attack 'fgsm' (eps=0.1) misclassified 93/100 images (93.0%). Below 0.98% threshold — will escalate attack next iteration.
[PLANNER | Iteration 1]:
**Analysis of Last Attack Result**

The last attack was an 'fgsm' with an epsilon value of 0.1, which resulted in a high misclassification rate of 93%. However, the model's predictions w

**History of Attempts**

This is the first iteration, so we have no previous attempts to analyze. We are starting fresh with a simple and fast attack.

**Choosing the Next Strategy**

Given the high success rate of the 'fgsm' attack but its relatively low quality (large differences between adversarial and clean predictions), I would

**Next Attack Type**

I recommend trying 'pgd' (Projected Gradient Descent) attack with a moderate epsilon value of 0.3. PGD is an extension of FGSM that projects the gradi

**Reasoning**

1. **Moderate epsilon**: By setting epsilon to 0.3 (an increase from 0.1), we're introducing stronger perturbations than before while still keeping it
2. **PGD is an extension of FGSM**: Since the 'fgsm' attack was successful, but introduced quite strong perturbations, PGD should provide a more targe
3. **Less destructive perturbations**: The projected gradient constraint in PGD may result in less visually obvious perturbations compared to FGSM, w

**Conclusion**
```

Results: Whitebox Attacks on Image Models

Dataset: ImageNet

Model Architecture	Accuracy on Benign Samples	Iterations	Initial Attack Used	Final Attack Used
ResNet-50	80.0%	2	FGSM	PGD
ViT	79.0%	2	FGSM	PGD
ResNet-50 AT	61.0%	1	PGD	PGD

Summary Table

					L-inf (stealth)	L2 (distortion)	Efficiency
					0.4349	1.5051	0.66
					0.4349	1.5051	0.66
					0.2500	87.7791	0.01
					0.1250	38.7862	0.03
					0.4349	1.5051	0.66
					0.0313	11.8437	0.06
					1.0000	53.4314	0.01
					0.4349	1.5051	0.66
					0.4349	1.5051	0.66
					0.4349	1.5051	0.66
8	carlin						
9	carlini_l2	0.00195	100.0%	0.4349	1.5051	0.66	

Final Verdict

Metric	Value
Target Type	Vision (ResNet-50 / ImageNet)
Total Iterations	10
Attack Successful	❌ NO
Best L-inf	0.0313 (Iteration 5 - fgsm)
Best L2	1.5051 (carlini_l2 consistently)
Best Efficiency Score	0.66 (carlini_l2)
Best Misclassification	100.0% (multiple iterations)

Efficiency Score
0.66
0.66
0.01
0.03
0.66
0.06
0.01
0.66
0.66

Results: Blackbox Attacks on Image Models

Dataset: ImageNet

Model Architecture	Accuracy on Benign Samples	Iterations	Initial Attack Used	Final Attack Used
ResNet-50	87.5%	2	Square	Square
ViT	79.0%	2	Square	Square
ResNet-50 AT	43.8%	3	Square	HSJ

Bla

Summary Table

Iter	Attack	Epsilon	Misclass%	L-inf	L2	Efficiency Score
0	hopskipjump	0.10000	100.0%	0.1189	19.5374	0.05
1	square	0.05000	25.0%	0.0500	15.1845	0.02
2	square	0.07500	100.0%	0.0750	27.1882	
3	square	0.05000	25.0%	0.0500	15.9412	
4	simba	0.05000	0.0%	0.0063	0.4999	
5	boundary	0.05000	100.0%	0.4164	20.0560	
6	hopskipjump	0.10000	75.0%	0.1137	18.9503	
7	hopskipjump	0.05000	100.0%	0.1142	18.8022	0.05
8	hopskipjump	0.05000	100.0%	0.1149	18.7172	0.05
9	hopskipjump	0.05000	100.0%	0.1161	18.7874	0.05

Best Performers by Metric

Metric	Best Value	Attack (Iteration)
Highest Misclassification	100.0%	hopskipjump (0,7,8,9
Lowest L-inf (stealthiest)	0.0063	simba (4)
Lowest L2 (smallest distortion)	0.4999	simba (4)
Highest Efficiency Score	0.05	hopskipjump, bound

Audio Data Models: Results Comparison

Dataset: Google Speech Command

Model	Baseline Accuracy	Attack Sequence	Success Rates/ MisClassification Error Rate	Iterations
ResNet18	99.7	DeepFool	100%	1
SimpleCNN	98	PGD	100%	1
VGGish	96.8	DeepFool	98%	1

Audio: VGGish

relative strengths is deepfool > carlini_l2 > pgd > fgsm

```
# MAE calculation
perturbation_mae = np.mean(np.abs(diff))

# SNR calculation
signal_power = np.mean(x_test ** 2)
noise_power = np.mean(diff ** 2)
snr = 10 * np.log10(signal_power / noise_power)
```

```
============================================================
ATTACK LOOP COMPLETE
============================================================

================================================================================
FINAL SUMMARY TABLE - Target: audio
================================================================================

Iter   Attack            Epsilon    Misclass%    Success    Composite
--------------------------------------------------------------------------------
1      deepfool          0.1        100.0%       0.1956
2      carlini_l2        0.1        95.9%        0.1956
3      pgd               0.1        91.8%        0.1635
4      fgsm              0.1        100.0%       0.1949
5      carlini_l2        0.1        100.0%       0.1949
6      deepfool          0.1        100.0%       0.1956
7      deepfool          0.05       100.0%       0.1956
8      deepfool          0.01       100.0%       0.1956
9      deepfool          0.01       100.0%       0.1956
10     pgd               0.5        100.0%       0.3683

================================================================================
DETAILED PERTURBATION METRICS
================================================================================

Iter   Attack            Misclass%    Perturbation Metric 1 Perturbation Metric 2
--------------------------------------------------------------------------------
1      deepfool          100.0       % MAE: 0.1038          SNR: 11.3dB
2      carlini_l2        95.9        % MAE: 0.0889          SNR: 20.7dB
3      pgd               91.8        % MAE: 0.1000          SNR: 20.0dB
4      fgsm              100.0       % MAE: 0.1038          SNR: 11.3dB
5      carlini_l2        100.0       % MAE: 0.1038          SNR: 11.3dB
6      deepfool          100.0       % MAE: 0.1038          SNR: 11.3dB
7      deepfool          100.0       % MAE: 0.1038          SNR: 11.3dB
8      deepfool          100.0       % MAE: 0.4086          SNR: 7.2dB

================================================================================
FINAL VERDICT
================================================================================
  Target Type           : audio
  Total Iterations      : 10
  Attack Successful     : ✅ YES
  Best Composite Score  : 0.1635 (Iteration 3)
  Best Misclassification: 91.8%
  Best Perturbation     : MAE=0.1000, SNR=20.0dB

================================================================================
```

Audio: Simple_CNN

relative strengths is deepfool > carlini_l2 > pgd > fgsm

```
============================================================
ATTACK LOOP COMPLETE
============================================================

================================================================================
FINAL SUMMARY TABLE - Target: audio
================================================================================

Iter   Attack            Epsilon    Misclass%    Success    Composite
--------------------------------------------------------------------------------
1      carlini_l2        0.1        100.0%       0.1107
2      deepfool          0.1        100.0%       0.1107
3      carlini_l2        0.1        100.0%       0.1318
4      pgd               0.1        97.3%        0.1318
5      carlini_l2        0.1        100.0%       0.1944
6      pgd               0.2        100.0%       0.1649
7      fgsm              0.1        100.0%       0.1107
8      deepfool          0.2        100.0%       0.1107
9      carlini_l2        0.3        100.0%       0.1107
10     deepfool          0.25       100.0%       0.1107

================================================================================
DETAILED PERTURBATION METRICS
================================================================================

Iter   Attack            Misclass%    Perturbation Metric 1 Perturbation Metric 2
--------------------------------------------------------------------------------
1      carlini_l2        100.0       % MAE: 0.0421          SNR: 22.1dB
2      deepfool          100.0       % MAE: 0.0785          SNR: 21.5dB
3      carlini_l2        100.0       % MAE: 0.1473          SNR: 15.9dB
4      pgd               97.3        % MAE: 0.0998          SNR: 20.0dB
5      carlini_l2        100.0       % MAE: 0.0421          SNR: 22.1dB
6      pgd               100.0       % MAE: 0.0421          SNR: 22.1dB

================================================================================
FINAL VERDICT
================================================================================
  Target Type           : audio
  Total Iterations      : 10
  Attack Successful     : ✅ YES
  Best Composite Score  : 0.1107 (Iteration 1)
  Best Misclassification: 100.0%
  Best Perturbation     : MAE=0.0421, SNR=22.1dB

================================================================================
```

Audio: Resnet18_A

```
================================================================================
FINAL SUMMARY TABLE - Target: audio
================================================================================

Iter   Attack            Epsilon    Misclass%    Success    Composite
--------------------------------------------------------------------------------
1      deepfool          0.1        100.0%       0.1358
2      deepfool          0.05       100.0%       0.1358
3      fgsm              0.1        98.0%        0.1612
4      carlini_l2        0.1        100.0%       0.1612
5      pgd               0.1        100.0%       0.1311
6      pgd               0.2        100.0%       0.1880
7      pgd               0.15       100.0%       0.1622
8      deepfool          0.05       100.0%       0.1358
9      carlini_l2        0.05       100.0%       0.1358
10     deepfool          0.1        100.0%       0.1358

================================================================================
DETAILED PERTURBATION METRICS
================================================================================

Iter   Attack            Misclass%    Perturbation Metric 1 Perturbation Metric 2
--------------------------------------------------------------------------------
1      deepfool          100.0       % MAE: 0.0539          SNR: 18.2dB
2      deepfool          100.0       % MAE: 0.0539          SNR: 18.2dB
3      fgsm              98.0        % MAE: 0.1000          SNR: 20.0dB
4      carlini_l2        100.0       % MAE: 0.0779          SNR: 21.6dB
5      pgd               100.0       % MAE: 0.1388          SNR: 16.3dB
6      pgd               100.0       % MAE: 0.1091          SNR: 18.5dB
7      pgd               100.0       % MAE: 0.0539          SNR: 18.2dB
8      deepfool          100.0       % MAE: 0.0539          SNR: 18.2dB

================================================================================
FINAL VERDICT
================================================================================
  Target Type           : audio
  Total Iterations      : 10
  Attack Successful     : ✅ YES
  Best Composite Score  : 0.1311 (Iteration 5)
  Best Misclassification: 100.0%
  Best Perturbation     : MAE=0.1388, SNR=16.3dB

================================================================================
```

Tabular Data Model: Results Comparison

Dataset: Irls

Model	Baseline Accuracy	Attack Sequence	Success Rates/ MisClassification Error Rate	Iterations
Random Forest	88.9	FGSM -> Boundary Attack	93.0% 100%	2
MLP	96.7	Boundary Attack	100%	1
Logistic Regression	91.1	FGSM->PGD->ZOO->DecisionTree->Boundary Attack	15.6% 31.1% FAILED-limited Batchsize Failed- Incompatibility with Logistic Regression 100%	5
Gradient Boost	93.3	PGD->ZOO->Boundary Attack->	Failed Failed 100%	3

Structured: MLP

The current ranking is:

`boundary_attack  > pgd > fgsm>  zoo > decision_tree`

ZOO and decision tree cannot perform a proper attack

Forget abou tht emisclassification rate

```
# Features changed calculation
feature_changes = np.abs(diff) > (epsilon / 2)
avg_features_changed = np.mean(np.sum(feature_changes, axis=1))
pct_features_changed = avg_features_changed / x_test.shape[1]

# L2 calculation
l2_per_sample = np.sqrt(np.sum(diff ** 2, axis=1))
perturbation_l2 = np.mean(l2_per_sample)
```

```
PS C:\Users\Jovin\Desktop\EE6039_FinalProj> python .\test.py
================================================================================

Iter   Attack             Epsilon    Misclass%    Success    Composite
--------------------------------------------------------------------------------
1      zoo                0.1        100.0%       0.2989
2      boundary_attack    0.1        100.0%       0.2989
3      decision_tree      0.05       100.0%       0.2989
4      fgsm               0.1        100.0%       0.2989
5      pgd                0.1        100.0%       0.2992
6      boundary_attack    0.05       100.0%       0.2992
7      fgsm               0.1        100.0%       0.2992
8      zoo                0.1        100.0%       0.2991
9      boundary_attack    0.1        100.0%       0.3002
10     boundary_attack    0.05       100.0%       0.3002

================================================================================
DETAILED PERTURBATION METRICS
================================================================================

Iter   Attack             Misclass%    Perturbation Metric 1 Perturbation Metric 2
--------------------------------------------------------------------------------
1      zoo                100.0       % FeatΔ: 99.4%         L2: 1.334
2      boundary_attack    100.0       % FeatΔ: 100.0%        L2: 1.323
3      decision_tree      100.0       % FeatΔ: 100.0%        L2: 1.322
4      fgsm               100.0       % FeatΔ: 100.0%        L2: 1.336

================================================================================
FINAL VERDICT
================================================================================
  Target Type           : structured
  Total Iterations      : 10
  Attack Successful     : ✅ YES
  Best Composite Score  : 0.2989 (Iteration 1)
  Best Misclassification: 100.0%
  Best Perturbation     : 99.4% features, L2=1.334

================================================================================
```

Structured: Random_

The current ranking is:

`boundary_attack > pgd > fgsm> zoo > decision_tree

```
================================================================================
FINAL SUMMARY TABLE - Target: structured
================================================================================

Iter   Attack             Epsilon    Misclass%    Success    Composite
--------------------------------------------------------------------------------
1      boundary_attack    0.1        100.0%       0.1772
2      pgd                0.05       100.0%       0.1772
3      zoo                0.1        100.0%       0.1772
4      decision_tree      0.05       100.0%       0.1772
5      fgsm               0.05       100.0%       0.1772
6      boundary_attack    0.1        100.0%       0.1902
7      fgsm               0.05       100.0%       0.1902
8      boundary_attack    0.05       100.0%       0.2213
9      zoo                0.05       100.0%       0.2213
10     zoo                0.1        100.0%       0.2213

================================================================================
DETAILED PERTURBATION METRICS
================================================================================

Iter   Attack             Misclass%    Perturbation Metric 1 Perturbation Metric 2
--------------------------------------------------------------------------------
1      boundary_attack    100.0       % FeatΔ: 61.1%         L2: 0.733
2      pgd                100.0       % FeatΔ: 63.9%         L2: 0.832
3      zoo                100.0       % FeatΔ: 76.7%         L2: 0.906

================================================================================
FINAL VERDICT
================================================================================
  Target Type           : structured
  Total Iterations      : 10
  Attack Successful     : ✅ YES
  Best Composite Score  : 0.1772 (Iteration 1)
  Best Misclassification: 100.0%
  Best Perturbation     : 61.1% features, L2=0.733

================================================================================
```

Structured: gradient_b

The current ranking is:

`boundary_attack  > pgd > fgsm>  zoo > decision_tree`

```
Iter   Attack             Epsilon    Misclass%    Success    Composite
--------------------------------------------------------------------------------
1      boundary_attack    0.1        100.0%       0.1080
2      boundary_attack    0.05       100.0%       0.1133
3      zoo                0.01       100.0%       0.1133
4      pgd                0.01       100.0%       0.1133
5      boundary_attack    0.01       100.0%       0.1267
6      fgsm               0.01       100.0%       0.1267
7      boundary_attack    0.01       100.0%       0.1230
8      boundary_attack    0.01       100.0%       0.1124
9      pgd                0.01       100.0%       0.1124
10     boundary_attack    0.01       100.0%       0.1275

================================================================================
DETAILED PERTURBATION METRICS
================================================================================

Iter   Attack             Misclass%    Perturbation Metric 1 Perturbation Metric 2
--------------------------------------------------------------------------------
1      boundary_attack    100.0       % FeatΔ: 33.3%         L2: 0.551
2      boundary_attack    100.0       % FeatΔ: 37.2%         L2: 0.518
3      zoo                100.0       % FeatΔ: 41.7%         L2: 0.579
4      pgd                100.0       % FeatΔ: 40.6%         L2: 0.558
5      boundary_attack    100.0       % FeatΔ: 38.3%         L2: 0.477
6      fgsm               100.0       % FeatΔ: 40.0%         L2: 0.634

================================================================================
FINAL VERDICT
================================================================================
  Target Type           : structured
  Total Iterations      : 10
  Attack Successful     : ✅ YES
  Best Composite Score  : 0.1080 (Iteration 1)
  Best Misclassification: 100.0%
  Best Perturbation     : 33.3% features, L2=0.551

================================================================================
```

Structured: Logistic Regr

The current ranking is:

`boundary_attack  > pgd > fgsm>  zoo > decision_tree`

```
================================================================================
FINAL SUMMARY TABLE - Target: structured
================================================================================

Iter   Attack             Epsilon    Misclass%    Success    Composite
--------------------------------------------------------------------------------
1      boundary_attack    0.1        100.0%       0.2150
2      zoo                0.1        13.3%        0.2150
3      fgsm               0.1        13.3%        0.7783
4      decision_tree      0.1        100.0%       0.7783
5      pgd                0.1        13.3%        0.7783
6      zoo                0.1        13.3%        0.7783
7      boundary_attack    0.1        13.3%        0.2177
8      zoo                0.1        13.3%        0.2177
9      decision_tree      0.1        13.3%        0.2177
10     fgsm               0.1        13.3%        0.7783

================================================================================
DETAILED PERTURBATION METRICS
================================================================================

Iter   Attack             Misclass%    Perturbation Metric 1 Perturbation Metric 2
--------------------------------------------------------------------------------
1      boundary_attack    100.0       % FeatΔ: 73.3%         L2: 0.911
2      zoo                13.3        % FeatΔ: 100.0%        L2: 0.200
3      fgsm               13.3        % FeatΔ: 100.0%        L2: 0.200
4      decision_tree      100.0       % FeatΔ: 74.4%         L2: 0.918
5      pgd                13.3        % FeatΔ: 100.0%        L2: 0.200

================================================================================
FINAL VERDICT
================================================================================
  Target Type           : structured
  Total Iterations      : 10
  Attack Successful     : ✅ YES
  Best Composite Score  : 0.2150 (Iteration 1)
  Best Misclassification: 100.0%
  Best Perturbation     : 73.3% features, L2=0.911

================================================================================
```

Results: LLMs Attacks (Limitation)

Most interesting one so far.

For Some reason every locally hosted LLM (we tried around 15-20 LLMs) had this same behavior.

- 0-2 iteration of prompts: Target LLM responds with,
  - ~~"I cannot respond to prompts of that nature"
- 2-5 iterations of prompts: target LLM just dont respond
  - We had a context checker to look for refusal from the LLM, where our agent saw no output as in no refusal and reported a successful attack, when the attack didnt even go through.
- 5> iterations: You could no longer use the LLM even for regular prompts.

Limitations cont'd

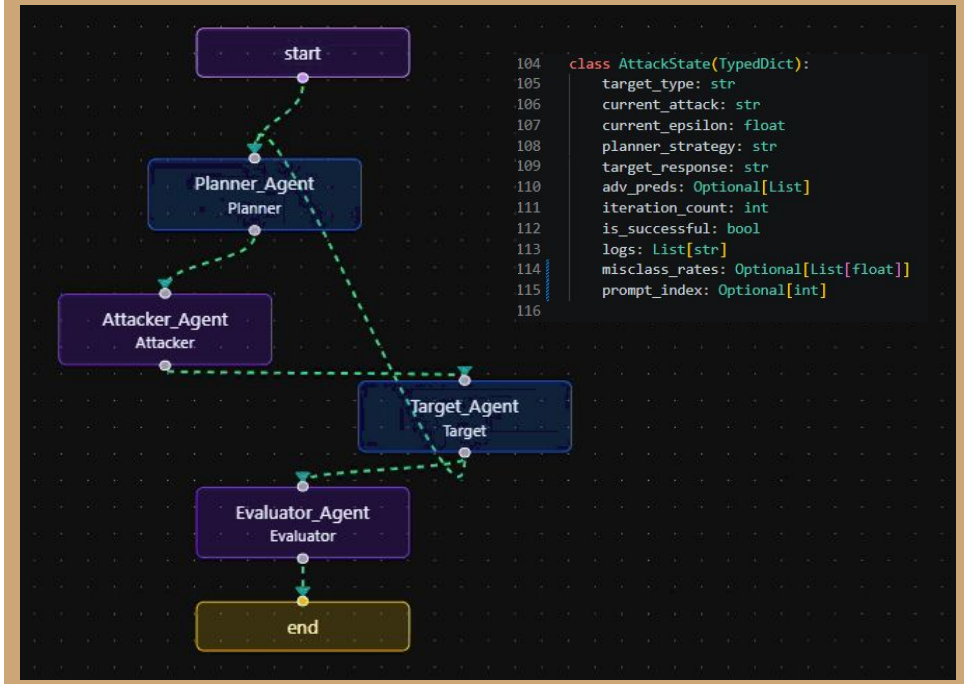
- Our experiments can only be run locally on our computers → many complex attacks (Adversarial Patching, Deepfool) cannot be computed in a reasonable time, especially for Vision attacks (works for some less dimensional data)
- As a result of having locally-dependent LLMs, we **cannot** test our project on proprietary LLMs (Gemini, ChatGPT, Claude).

Next Steps

- Add support for LLM attacks.
- Adding a Defender Agent (Adversarial Red-Teaming)

As a future objective, where we can develop an LLM to optimize a defense model based on the improved versions of the original LLM.

- Test in BlackBox Scenarios

QUESTIONS?

Thank you!